FUSION 2026

Problem Statements — IOT

IOT-01 — Predictive Failure Detection for Smart Grid Substations

Organization: Siemens

Problem Statement : Substation equipment failures cause cascading power outages. Sensor data (temperature, vibration, load, voltage fluctuation) is collected but rarely used to predict failure before it happens.

Expected Solution : Using a public or simulated substation sensor-time-series dataset, build a model that predicts imminent equipment failure, flags the specific sensor readings driving the prediction, and visualizes risk across a mock substation layout.

Dataset / Reference
https://www.kaggle.com/datasets/zara2099/smart-grid-terminal-fault-dataset
https://www.kaggle.com/datasets/shashwatwork/failure-analysis-in-power-transformers-dataset

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IOT-02 — Low-Power Anomaly Detection for Precision Irrigation Networks

Organization: John Deere

Problem Statement : Battery-powered soil-moisture and irrigation sensors deployed across large farms must detect anomalies (leaks, sensor failure, irrigation malfunction) locally, since constant cloud transmission drains battery life too fast.

Expected Solution : Design and demo an edge-friendly anomaly-detection approach (e.g., a lightweight model runnable on a microcontroller-class simulator) that flags irrigation anomalies from a sample sensor dataset while minimizing data sent to the cloud, and show the power/bandwidth trade-off.

Dataset / Reference
https://iotdataset.com/data/smart-agriculture-plant-health-iot-sensors

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IOT-03 — Real-Time Cold Chain Integrity Monitoring for Vaccine Logistics

Organization: UNICEF

Problem Statement : Vaccines require an unbroken cold chain from manufacturer to clinic. A temperature excursion during transit can silently spoil a shipment, and current tracking often only reveals the failure after delivery.

Expected Solution : Build a dashboard that ingests simulated temperature/humidity/GPS sensor data from a shipment, detects an excursion in near real-time, estimates remaining vaccine viability based on the excursion severity/duration, and alerts the nearest checkpoint before delivery.

Dataset / Reference

https://www.kaggle.com/datasets/skarin/cold-chain-shipment-silent-failure-dataset
https://www.kaggle.com/datasets/manankhanna0/vaccine-distribution-with-temperature-logging

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IOT-04 — Swarm Coordination for Warehouse Robot Fleets

Organization: Amazon Robotics

Problem Statement : Warehouse robot fleets must dynamically re-route around congestion, equipment failure, and priority order changes without a central bottleneck controller re-planning everything from scratch.

Expected Solution : Build a simulated warehouse grid with multiple robot agents that coordinate path-finding and task allocation in real time, demonstrating graceful re-routing when a path is blocked or a robot "fails," visualized on a 2D grid.

Dataset / Reference

No dataset needed
Benchmark: https://movingai.com/benchmarks/mapf.html

IOT-05 — Occupancy-Aware Smart Building Energy Optimization

Organization: Schneider Electric

Problem Statement : Commercial buildings waste significant energy heating/cooling/lighting spaces based on fixed schedules rather than actual occupancy, but installing dense occupancy sensors is expensive**.**

Expected Solution : Using a sample dataset combining sparse occupancy sensors, HVAC schedules, and weather data, build a model that infers room-level occupancy patterns and recommends a dynamic HVAC/lighting schedule, quantifying estimated energy savings versus the fixed-schedule baseline.. .

Dataset / Reference

Occupancy Detection - UCI Machine Learning Repository
https://github.com/buds-lab/building-data-genome-project-2